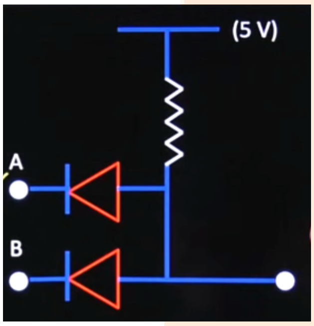
JEE-Main-28-01-2026 (Memory Based)
[EVENING SHIFT]
Physics

Question: Which of the following can not be measured?
Options:
(a) Resistance
(b) Potential
(c) Potential difference
(d) Displacement current
Answer: (b)

Question: For the circuit given below, identify the logic gate.



Options:
(a) AND
(b) OR
(c) NAND
(d) NOR
Answer: (a)

Question: Object is placed at distance 30 cm from lens given below, then distance of image from lens is (R_1 = 10 cm, R_2 = 20 cm)
Options:
(a) 36 cm
(b) 24 cm
(c) 20 cm
(d) 30 cm
Answer: (b)

Question: If position vector is given as $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and if its signs are reversed then which of the following physical quantity remains unaffected?
Options:
(a) Acceleration

(b) Velocity
(c) Displacement
(d) Torque
Answer: (d)

Question: If the mass number of nucleus is α, its radius is R_α, and another mass number is β then its radius is R_β; then R_α/R_β = ? [Given β = 8 α]
Options:
(a) 8
(b) 1/2
(c) 1
(d) 2
Answer: (b)

Question: Which of following physical quantity is not measurable?
Options:
(a) Displacement
(b) Voltage
(c) Voltage difference
(d) Acceleration
Answer: (b)

Question: Two light sources of X_1 and X_2 are used for YDSE with slit distance 2.25 mm and distance between the slits and screen is 1.5 m. Then the distance from central maxima for which minima of both wavelength coincide?
Options:
(a) 1.65 mm
(b) 1.20 mm
(c) 1.30 mm
(d) 1.40 mm
Answer: (a)

Question: A beam of power 2 μW is hitting a metal surface. Beam contains photons of wavelength 662 nm. Find number of photons striking per second.
Options:
(a) 2×10^{14}
(b) 6.67×10^{12}
(c) 4×10^{11}
(d) 3.2×10^{13}
Answer: (b)

Question: Mean free path of gas particles of diameter 5 Å at temperature and pressure of 41℃ and 1.38×10^5 Pa.
Options:
(a) 14.14 nm
(b) 20 nm
(c) 28.28 nm
(d) 10 nm
Answer: (c)

Question: Find the percentage error in K, where $T = 2\pi\sqrt{\frac{m}{K}}$. Given that 60 oscillations completes in 50 second. Time resolution is 2 second, m = 10 grand $\Delta m = \pm 10$ mg.

Options:

(a) 8%

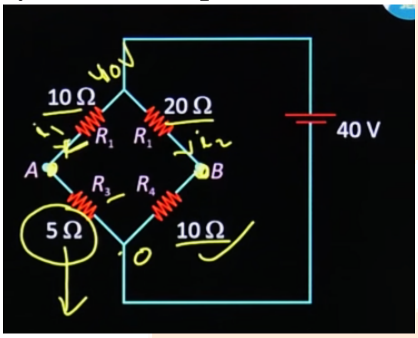
(b) 9%

(c) 9.1%

(d) 8.1%

Answer: (d)

Question: In a balanced Wheatstone bridge $R_2R_3 : R_1R_4$. Because of heating R_3 increases by 20%. Then potential difference across A and B becomes



Options:

(a) 1.50 V

(b) 2.40 V

(c) 1.67 V

(d) 3.60 V

Answer: (c)

Question: Match the two columns and choose the correct option.

	Column - I		Column - II
(a)	Coefficient of viscosity	(p)	$[ML^0T^{-2}]$
(b)	Surface tension	(q)	$[ML^{-1}T^{-2}]$
(c)	Pressure	(r)	$[ML^2T^{-2}]$
(d)	Work	(s)	$[ML^{-1}T^{-1}]$

Options:

(a) (a)-(p), (b)-(q), (c)-(r), (d)-(s)

(b) (a)-(s), (b)-(p), (c)-(q), (d)-(r)

(c) (a)-(q), (b)-(s), (c)-(p), (d)-(r)

(d) (a)-(p), (b)-(q), (c)-(s), (d)-(r)

Answer: (b)

Question: Given that $v=\sqrt{\frac{Y}{P}}$. **Find the maximum % error in v. Given that** $\frac{\Delta Y}{Y}\times 100=1\%$ and $\frac{\Delta P}{P}\times 100=0.5\%$

Options:

(a) 3/2 %

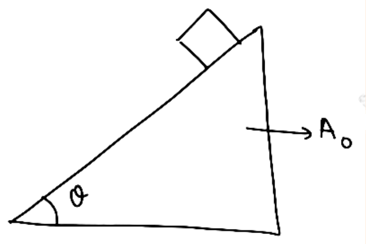
(b) 3/4 %

(c) 1 %

(d) 1/2 %

Answer: (b)

Question: A block of mass m is placed on a smooth inclined plane making an angle θ with the horizontal. The inclined plane itself is accelerating horizontally with constant acceleration A0. Find the time taken by the block to reach the bottom of the incline (length = L)



Options:

(a) $\sqrt{\frac{2l}{A_0\cos\theta+g\sin\theta}}$

(b) $\sqrt{\frac{2l}{A_0\sin\theta+g\cos\theta}}$

(c) $\sqrt{\frac{2l}{(A_0+g)\sin\theta}}$

(d) $\sqrt{\frac{2l}{gsin\theta}}$

Answer: (a)

Question: Consider the following electromagnetic waves Wavelength of A = 400 nm, Frequency of B = 10^{16} sec^{-1}, Wave number of C = 10^4 cm^{-1} order of energies is:

Options:

(a) A > B > C

(b) B > A > C

(c) B > C > A

(d) C > A > B

Answer: (b)

Question: Two turning forks A & B produce 8 beats in 2 seconds. When 'A' waxed, the number of beats becomes 4 in 2 seconds. Find frequency of 'A' if frequency of 'B' is 380 Hz?

Options:
(a) 376 Hz
(b) 384 Hz
(c) 380 Hz
(d) None
Answer: (b)

JEE-Main-28-01-2026 (Memory Based)
[EVENING SHIFT]
Chemistry

Question: The plot of $\log_{10}$ K vs $\frac{1}{T}$ gives a straight line. The intercept and slope respectively are

Options:

(a) $c = log\,A,\ m =- \frac{E_a}{2.303R}$

(b) $c =- \frac{E_a}{2.303R'} m = logA$

(c) $c =- \ log\,A,\ m =- \frac{E_a}{2.303R}$

(d) $c = log\,A,\ m = \frac{E_a}{2.303R}$

Answer: (a)

Question: Consider the following electromagnetic waves
Wavelength of A = 400nm
Frequency of B = 10^{16} sec^{-1}
Wave number of C = 10^4 cm^{-1}
Order of energies is:

Options:

(a) A > B > C
(b) B > A > C
(c) B > C > A
(d) C > A > B

Answer: (b)

Question: Which of the following order is correct.

Options:

(a) HF > HI > HBr > HCl (Boiling point)
(b) HF > HI > HBR > HCL (Melting point)
(c) HI > HF > HBr > HCl (Boiling point)
(d) HI > HBr > HF > HCl (Melting point)

Answer: (a)

Question: Consider a reaction A ⇌ B. At 'T' K, the equilibrium concentration of A and B are 0.3 M and 0.315 M. Now, 0.1 mol of A is added to the flask of 1 L, then equilibrium constant and equilibrium concentration of B are

Options:

(a) 1.05, 0.35 M
(b) 0.95, 0.37 M
(c) 1.05, 0.37 M
(d) 0.95, 0.35 M

Answer: (c)

Question: The sum of valence e^- in element with most and least metallic character among the following is:
Na, P, Cl, S, O and F
Options:
Answer: (8)

Question: Match the isostructural species

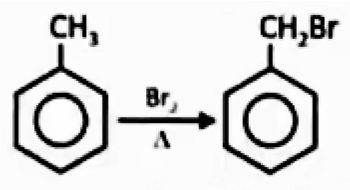
Column-I	Column-II
a) XeO_3	p) BrF_5
b) XeF_2	q) NH_3
c) XeO_2F_2	r) I_3^-
d) $XeOF_4$	s) SF_4

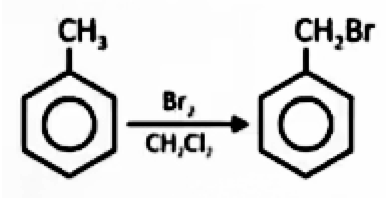
Options:
(a) a – q, b – r, c – s, d - p
(b) a – p, b – q, c – s, d - p
(c) a – q, b – r, c – p, d - s
(d) a – p, b – q, c – r, d - s
Answer: (a)

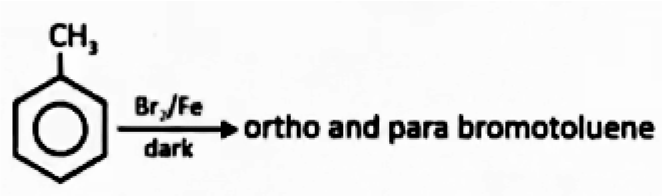
Question: Diamagnetic species among the following complexes is
Options:
(a) $[MnBr_4]^{2-}$
(b) $[Cu(H_2O)_6]^{2+}$
(c) $[Ni(CN)_4]^{2-}$
(d) $[Ni(H_2O)_6]^{2+}$
Answer: (c)

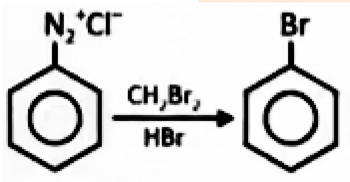
Question: Correct statement about -NO_2 group is -NO_2 is
A) Ring deactivating group in electrophilic substitution
B) Ring activating group in electrophilic substitution
C) Activating for aromatic nucleophilic substitution in Aryl halides
D) Deactivating for aromatic nucleophilic substitution in Aryl halides.
Options:
(a) A, C are correct statement
(b) B, D are correct
(c) A, D are correct
(d) B, C are correct
Answer: (a)

Question: The major organic product of which of the following reaction is incorrectly represented?
Options:

(a) 

(b) 

(c) 

(d) 

Answer: (a, c, d)

Question: Which of the following compounds on reacting with Hinsberg Reagent forms an alkali insoluble product.
A) Ethanamine
B) N-Methylaniline
C) N-Ethyl-N-Methylaniline
D) N-Methylthionine
E) N-Phenylaniline
F) Aniline
Options:
(a) A, C, D, E only
(b) B, C, D & E only
(c) B, D and E only
(d) A, C, F only
Answer: (c)

Question: In 'S' estimation, 0.314 g of organic compound gave 0.4813 g fo barium sulphide. What is % of 'S' in organic compound?
(Report to nearest integer).
Options:
Answer: (21)

Question: Among Sc^{3+}, Cr^{2+}, Mn^{3+}, Co^{3+} number of isoelectronic species are 'n'. 'n' moles of AgCl is obtained upon reaction with excess of $AgNO_3$ with 1 mol of $Co(en)_2NH_3Cl_3$.

Number of t_{2g} electrons in the complex are
Options:
Answer: (6)

Question: An alpha particle and proton are accelerated in a discharge tube under same potential difference of 200 KeV. The debroglie wavelength of proton is $x\sqrt{2}$ times of de Broglie wavelength of α-particle. The value of x is
Options:
Answer: (2)

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[EVENING SHIFT]
Maths

Question: $\frac{6}{3^{26}} + \frac{10}{3^{25}} + \frac{10.2}{3^{24}} + \frac{10.2^2}{3^{23}} + ... + \frac{10.2^{24}}{3}$ **is equal to**

Options:

(a) 2^{26}

(b) 2^{25}

(c) 3^{26}

(d) 3^{25}

Answer: (a)

Question: The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{12(3+[x])}{3+[\sin x]+[\cos x]}dx$ **is equal to**

Options:

(a) $3 + 10\pi$

(b) $11\pi + 4$

(c) $10\pi + 2$

(d) $11\pi + 2$

Answer: (d)

Question: By the principal of inverse trigonometric function, the value of
$\tan\left(2\sin^{-1}\left(\frac{2}{\sqrt{13}}\right) - 2\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)\right)$ **is equal to**

Options:

(a) $\frac{33}{56}$

(b) $\frac{31}{55}$

(c) $\frac{32}{59}$

(d) $\frac{38}{55}$

Answer: (a)

Question: Let a triangle ABC such that A ≡ (0, 0) and vertices b and C lie on the parabola $y^2 = 8x$ such that $\left(\frac{7}{3}, \frac{4}{3}\right)$ **is the centroid of the ΔABC then $(BC)^2$ is equal to**

Options:
(a) 90
(b) 120
(c) 150
(d) 110
Answer: (b)

Question: Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ **and B be 2 × 2 matrix such that A^{100} = 100B + I, then sum of all elements of B^{100} is**
Answer: $B^{100} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Question: If the arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}$ is $\frac{5}{16}$ **and a, 4, α , β are in increasing A.P. then both the roots of the equation $ax^2 - ax + 2(\alpha - 2b) = 0$ lie between**
Options:
(a) (-3, 0)
(b) (-2, 3)
(c) (0, 3)
(d) (-3, 1)
Answer: (b)

Question: The range of f(x) = sgn(sinx) + sgn(cosx) + sgn(tanx) + sgn(cotx), $x \neq \frac{n\pi}{2}$, **n $\in$ I, where** $\operatorname{sgn}(t) = \begin{cases} 1, & t > 0 \\ -1 & t < 0 \\ 0, & t = 0 \end{cases}$
Options:
(a) {4, -4, 2, -2}
(b) {-2, 0, 4}
(c) {4, -4, 0, -2}
(d) {2, -2, 0, 4, -4}
Answer: (b)

Question: Statement I : $25^{13} + 20^{13} + 8^{13} + 3^{13}$ is divisible by 7
Statement II : The integral value of $(7 + 4\sqrt{3})^{25}$ is an odd number
Options:
(a) Neither statements are correct
(b) Only statement I is correct
(c) Only statement II is correct
(d) Both the statements are correct
Answer: (d)

Question: Let y = y(x) be the solution of the differential equation $x\frac{dy}{dx} - y = x^2 \cot x, x \in (0, \pi)$. **If** $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then $6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right) is$

Options:

(a) 2π

(b) -3π

(c) -π

(d) π

Answer: (c)

Question: Statement I : The function F defined from R → R $F(x) = \frac{x}{1+|x|}$ **is one-one**

Statement II : The function F defined from R → R $F(x) = \frac{x^2+4x-30}{x^2-8x+18}$ **is may-are**

Options:

(a) Statement I is correct but Statement II is not correct

(b) Statement I and Statement II both are correct

(c) Statement I is incorrect but Statement II is correct

(d) Both Statement are incorrect

Answer: (b)

Question: Let $f(x) = \lim_{\theta \to 0} \frac{\cos \pi x - \left(x^{2/\theta}\right) \sin(x-1)}{1 + \left(x^{2/\theta}\right) \sin(x-1)}$, **x** $\in$ **R. Then which of the following is correct?**

Options:

(a) f is continuous at x = 1 and f(1) = -1

(b) f is discontinuous at x = -1 and f(1) = -1

(c) f is continuous at x = 1 and f(1) = 1

(d) f is discontinuous at x = 1 and f(1) = 1

Answer: (a)

Question: Ellipse $\frac{x^2}{144} + \frac{y^2}{169} = 1$ **and hyperbola** $\frac{x^2}{16} - \frac{y^2}{\lambda^2} = -1$ **have sam focus and e and L denotes the eccentricity and length of latus rectum of hyperbola then 24(e + l) is**

Answer: (296)

Question: An ellipse has centre at (1, -2) and one of the focus at (3, -2) and one vertex at (5, -2), then the length of its latus rectum is

Answer: (6)

Question: Consider the data:

x:	$4k$	$\frac{30}{7}k$	$\frac{32}{7}k$	$\frac{34}{7}k$	$\frac{36}{7}k$	$\frac{38}{7}k$	$\frac{40}{7}k$	$6k$
$p(x)$:	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$

If $E(x) = \frac{263}{15}$, **then P(x < 20) is equal to**

Options:

(a) $\frac{1}{15}$

(b) $\frac{8}{15}$

(c) $\frac{4}{15}$

(d) $\frac{14}{15}$

Answer: (d)

Question: Let Q be the image of the point P(3, 2, 1) in the line $\frac{x-1}{1} = \frac{y}{2} = \frac{z-1}{1}$, **then the distance of Q from the line** $\frac{x-9}{3} = \frac{y-9}{2} = \frac{3-5}{-2}$ **is**

Options:

(a) 3

(b) 4

(c) 5

(d) 7

Answer: (d)